Geotemporal Hydrodynamics

The Unified Fluid Theory

T. Abram

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Abstract

Geotemporal Hydrodynamics (GTH) proposes that spacetime is a macroscopic quantum superfluid rather than an empty geometric manifold. The framework is motivated by the vacuum-energy inconsistency between Quantum Field Theory and standard cosmology, and models the vacuum as a 5D viscoelastic substrate with order parameter $\Psi = \sqrt{\rho}e^{i\theta}$. In this picture, particles are not point objects but stable topological defects (GeoKnots), while gravity emerges as the long-wavelength acoustic response of the substrate.

Starting from a covariant hydrodynamic action augmented by Chern–Simons and Wess–Zumino–Witten sectors, the manuscript derives coupled continuity–Euler dynamics, an effective acoustic metric, and a constitutive map to a Poisson-limit gravitational equation. A closed parameterization is presented for effective gravitational coupling and defect-energy mass scaling, including explicit dependence on substrate quantities $(m_\psi, \rho_0, c_s, \tau, \eta)$. At galactic scales, the theory predicts wake-mediated transport and inverse-cascade contributions to rotation support; at compact-object scales, it predicts frequency-dependent viscoelastic suppression consistent with ringdown-fidelity constraints.

The theory is framed as falsifiable through three near-term channels: rotation-curve residual tests (SPARC), ringdown dispersion/damping constraints (LIGO/Virgo/KAGRA), and analog superfluid-vortex experiments that probe topological stability and transport signatures.

Axiomatic Executive Summary

To establish falsifiability and rigorous scale-invariance goals, this treatise develops a mathematical program for the following phenomena as testable hydrodynamic constructions requiring full derivation and validation:

I. **The Grand Unification of Newton's Constant** (G)
Gravity is not a fundamental force, but an induced acoustic pressure coupling. A proposed effective expression for Newton's Gravitational Constant is written in terms of the speed of sound (c_s), the reduced Planck constant $(\hbar)$, and the squared mass of the substrate scalar mode (m_ψ):

$$G_{eff} = \frac{3\pi\hbar c_s}{4m_\psi^2}$$

This recovers the correct $\mathrm{m}^3\mathrm{kg}^{-1}\mathrm{s}^{-2}$ dimensionality without auxiliary scaling parameters.

II. **The Acoustic Derivation of Fundamental Mass** (m_e)
Rest mass is not an intrinsic property; it is the localized Gross-Pitaevskii kinetic bending energy of the substrate. By modeling the Electron as a fundamental Trefoil Knot (3_1) bound by a Compton-resonant phase closure, the 0.511 MeV scale is modeled from the background fluid line-tension $(\mathcal{T}_{vacuum})$:

$$m_e = \sqrt{\frac{\rho_0 h^3}{4\pi c^3 m_\psi^2}\ln(\Lambda)}$$

III. **The Topological Derivation of the Pauli Exclusion Principle**
Vortices in standard superfluids are bosonic. However, by augmenting the 5D Master Action with a Wess-Zumino-Witten (WZW) topological framing term, the Finkelstein-Rubinstein theorem[1] shows that a 3_1 fluid knot acquires a -1 phase shift under a 2π spatial rotation. Fluid knots then carry half-integer spin in this construction, giving a mechanism consistent with Fermi-Dirac statistics and the Pauli Exclusion Principle.

IV. **Color Confinement as Dirichlet Boundary Mechanics**
Quarks are topologically identified as open fractional vortex braids. In a pressurized superfluid substrate, an open vortex filament requires infinite hydrodynamic strain energy at its severed boundaries. To prevent infinite energy bleed into the bulk, Dirichlet boundary conditions physically force the exposed ends to link into mutually sustaining Borromean geometries (Protons/Neutrons). Color Confinement is a thermodynamically motivated closure condition, requiring no virtual gluon exchange.

V. **Dark Matter as the Inverse Energy Cascade**
Standard models assume turbulent energy only cascades downward. Using contemporary computational fluid-dynamics analyses, GTH models that at the galactic scale $(Re \gg 1)$, microscopic quantum strain fields self-organize via the Inverse Energy Cascade. This upward transfer of kinetic energy generates a macroscopic, rotating fluid wake that yields a βr effective term in flat rotation-curve fits without requiring collisionless particulate Dark Matter.

VI. **Viscoelastic Immunity and GW250114**
A fluid-vacuum model must preserve high-frequency ringdown structure. The formal requirement is not just small viscosity, but frequency-dependent effective dissipation suppression in

the measured ringdown band. In the near-horizon merger regime, the substrate dynamics are governed by Israel-Stewart relaxation:

$$\tau \Delta^{\alpha}_{\mu} \Delta^{\beta}_{\nu} \dot{\Pi}_{\alpha\beta} + \Pi_{\mu\nu} = -2\eta\sigma_{\mu\nu}$$

In the relativistic limit $\omega\tau \gg 1$, effective viscous dissipation channels collapse as $\eta_{eff} \sim \eta/(1 + (\omega\tau)^2)$, and the medium responds elastically rather than diffusively. This preserves Kerr-like quasinormal-mode fidelity in GW250114 while remaining consistent with the same substrate dynamics used at larger scales. The key empirical point is that the same constitutive closure must simultaneously satisfy compact-object ringdown constraints and supercritical galactic transport signatures.

VII. **The Empirical Bridge (Macroscopic Isomorphisms)**
GTH closes theory-to-laboratory continuity by mapping the Abram-Paschen coupling to ultrasonic plasma-guiding analogs: imposed acoustic density gradients in a fluid steer charged plasma trajectories in structural correspondence with vacuum-pressure gradients steering effective charge transport in the substrate sector. The correspondence is used as a constrained mechanism analogy, not as a one-step proof of full quantum-gravitational identity; its value is to provide controllable experiments that validate transport motifs, response delays, and gradient-guided current routing under known boundary conditions.

Assumptions, Validity Domain, and Limitations

Assumption Set A (Continuum Closure). The substrate sector is treated as an effective continuum with well-defined coarse-grained fields $(\rho, \theta, u^{\mu}, \Pi_{\mu\nu})$ over scales larger than core-healing microstructure.

Assumption Set B (Topological Stability Window). Defect classes are assumed metastable over observational timescales in subcritical regimes, with reconnection rates exponentially suppressed relative to dynamical times.

Assumption Set C (Constitutive Causality). Dissipative dynamics are modeled with second-order causal relaxation (Israel–Stewart class), excluding first-order acausal truncations.

Validity Domain. Claims are intended for: (i) laminar/subcritical compact systems, (ii) supercritical galactic transport sectors, and (iii) high-frequency merger propagation where $\omega\tau \gg 1$. Extrapolation outside these regimes is explicitly non-assertive.

Primary Limitations. The framework depends on calibration of $(m_{\psi}, \xi, \tau, \eta)$ and on inference quality of wake geometry from finite observational resolution. The theory is therefore constrained by both constitutive assumptions and data-systematics floors.

Notation Key (Disambiguation). η_{RF} denotes RF conversion efficiency, while η and ζ denote shear and bulk viscosities in the constitutive sector. Unless explicitly marked as τ_{KZ} in critical-dynamics contexts, τ denotes Israel–Stewart relaxation time. The symbol θ denotes substrate phase, while Θ denotes the fluid expansion scalar. The density symbol ρ_0 denotes the background substrate density and is also used as the wake-medium density in drag-sector equations.

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Part I

Foundations of the Fluid Vacuum

1

The 5D Non-Newtonian Superfluid Substrate

1.1 The Ontological Shift: The Substance of Spacetime

In orthodox formulations, the vacuum appears as an unresolved tension: General Relativity models it as an empty geometric manifold, while Quantum Field Theory models it as a highly energetic sea of fluctuations.

Geotemporal Hydrodynamics (GTH) eliminates this paradox by redefining the ontological nature of spacetime. The vacuum is not an empty geometric arena; it is a physical, macroscopic substance—a 5D non-Newtonian superfluid substrate where the 4D spacetime metric is an emergent acoustic phenomenon.[2–4]

The macroscopic quantum state of this medium is described by the order parameter $\Psi = \sqrt{\rho}e^{i\theta}$, where ρ is the vacuum density and the phase gradient dictates the irrotational superfluid velocity $\mathbf{v}_s = \frac{\hbar}{m_\psi}\nabla\theta$. Standard Model particles are not zero-dimensional points; they are persistent, quantized topological defects (GeoKnots) sustained by the fluid's line tension. The metric of General Relativity ($g_{\mu\nu}$) is not fundamental, but modeled as emergent—an acoustic geometry dictated by the propagation of phonons through the substrate.

1.2 The Grand Unification of Newton's Constant (G_{eff})

In classical physics, Newton's Gravitational Constant (G) is usually treated as an empirical input. In GTH, gravity is modeled as an emergent acoustic pressure coupling generated by phase–density response of the substrate, rather than as a fundamental geometric primitive. The derivation strategy is to (i) linearize the substrate equations around a stationary background, (ii) identify the source coupling of defect density into the phase potential, and (iii) match the static limit to Poisson form to extract an effective coupling constant. This recasts G_{eff} from a stand-alone postulate into a constitutive quantity in the model class while keeping explicit dependence on calibration-sensitive parameters visible for falsification.

Applying the principles of induced gravity, the effective gravitational coupling between two mass-displacing acoustic sinks is defined by the fluid's speed of sound (c_s), the reduced Planck constant ($\hbar$), and the ultraviolet cutoff scale of the medium (Λ_{UV}):

$$G_{eff} = \frac{3\pi c_s^3}{2\hbar\Lambda_{UV}^2}$$

In a superfluid, the physical UV cutoff is strictly bounded by the intrinsic healing length (coherence length ξ), where $\Lambda_{UV} = 1/\xi$. Substituting the Gross-Pitaevskii relationship for coherence length ($\xi = \frac{\hbar}{\sqrt{2}m_\psi c_s}$), we isolate the exact fundamental derivation of the gravitational constant:

$$G_{eff} = \frac{3\pi\hbar c_s}{4m_\psi^2}$$

This unification formula requires no auxiliary scaling parameters. Dimensional analysis resolves to the units $m^3kg^{-1}s^{-2}$. Within the stated closure assumptions, Newton's Constant is modeled as a derived macroscopic ratio governed by the wave speed in the vacuum and the squared scalar-sector mass (m_ψ).

1.3 The Acoustic Mass Unification (m_e)

Orthodox physics treats the mass of the electron $(0.511$ MeV$)$ as an empirical free parameter. In GTH, mass is the integrated kinetic bending energy of the substrate required to sustain a specific topological phase defect against the background vacuum tension $(\mathcal{T}_{vacuum})$.

For a continuous fluid vortex to remain kinetically stable without dissipating, the quantum phase θ must satisfy the Bohr-Sommerfeld quantization condition. The physical circumference of the vortex tube must equal exactly one resonant standing wavelength—physically identical to the Compton Wavelength (λ_C). Therefore, the geometric arc length of the foundational 3_1 Trefoil knot (the Electron) is strictly $L_{knot} = \frac{h}{m_e c}$.

The base line-tension of the substrate is defined by its bulk density (ρ_0) and the quantum of circulation $(\kappa = h/m_\psi)$:

$$\mathcal{T}_{vacuum} = \frac{\rho_0 h^2}{4\pi m_\psi^2} \ln(\Lambda)$$

Equating the topological rest-mass energy $(m_e c^2)$ to the product of the fluid's topological line tension and the resonant arc length yields the Grand Unification formula for fundamental mass:

$$m_e = \sqrt{\frac{\rho_0 h^3}{4\pi c^3 m_\psi^2} \ln(\Lambda)}$$

The mass of the electron is modeled as a function of the vacuum density (ρ_0) and scalar mass scale (m_ψ). Under calibrated parameter choices, this localized fluid-strain estimate lands at the 0.511 MeV scale.

1.4 Dark Energy as Boundary Thermodynamics

The accelerated expansion of the universe is currently attributed to a mysterious "Dark Energy" producing an equation of state $w = -1$[5]. In GTH, because spacetime is a 5D fluid manifold, it exhibits classical boundary thermodynamics.

To expand the surface area of the universe, work must be done against the boundary surface tension (γ) of the bulk fluid $(dW = \gamma dA)$. This macroscopic fluid mechanic creates a uniform, negative Quantum Zero-Point Pressure. This boundary surface tension physically enforces the $w = -1$ equation of state, driving the exponential expansion of the metric without violating the conservation of energy.

In this manuscript, Eqs. (1.2) and (1.3) are not treated as isolated identities. They are required to emerge from the long-wavelength reduction chain in Eqs. (11.2.1)–(11.2.2) and remain consistent under shared-parameter, cross-channel constraints.

The same standard governs the boundary-work dark-energy channel: it is retained only when cross-channel ranking, predictive transfer, and rejection-criteria thresholds remain stable under predeclared conditions.

2

The Master Action & Covariant Formalism

2.1 The 5D Bulk Lagrangian

To establish a formal mathematical foundation for Geotemporal Hydrodynamics, we construct a Master Action (S_{GTH}) from which particle kinematics, emergent gravitational metrics, and topological constraints are derived.

The substrate operates on a 5-dimensional bulk manifold $\mathcal{M}_5$ with coordinates $X^A = (x^\mu, \tau)$, where τ represents the viscoelastic relaxation dimension. The base covariant action integrates the 5D Ricci curvature, the Gross-Pitaevskii superfluid dynamics, and the boundary surface tension:

$$\mathcal{S}_{base} = \int_{\mathcal{M}_5} d^5X \sqrt{-g_{(5)}} \left[\frac{1}{2} g^{AB} (D_A\Psi)^\dagger (D_B\Psi) - V(|\Psi|^2) \right] + \int_{\partial\mathcal{M}} d^4x \sqrt{-g_{(4)}} \gamma(\Psi)$$

Where the symmetry-breaking potential $V(|\Psi|^2) = -\mu^2|\Psi|^2 + \frac{\lambda}{2}|\Psi|^4$ establishes the vacuum expectation value (VEV) and the baseline density of the universe.

2.2 The Chern-Simons Helicity Constraint ($\mathcal{L}_{CS}$)

If particles are topological fluid vortices (GeoKnots), the action must mathematically forbid these knots from arbitrarily unspooling and dissipating into the bulk. We rigorously enforce this by augmenting the Lagrangian with a Chern-Simons topological boundary constraint:

$$\mathcal{L}_{CS} = \frac{k}{4\pi} \epsilon^{\mu\nu\rho\sigma} A^{eff}_\mu B^{eff}_{\nu\rho\sigma}$$

This term explicitly couples the fluid's velocity to its intrinsic helicity. It mathematically dictates that any localized "twisting" of the fluid's phase cannot be continuously deformed into a smooth vacuum state without expending infinite energy. The integer k strictly enforces the quantization of circulation, acting as an unbreakable geometric padlock that guarantees the immortality of the proton and the stability of atomic matter.

2.3 The WZW Action and the Pauli Exclusion Principle

A historic critique of modeling the vacuum as a superfluid is that macroscopic vortices are inherently bosonic, whereas fundamental matter (electrons, quarks) are spin-$1/2$ fermions subject to the Pauli Exclusion Principle.

To resolve this, the GTH Master Action integrates a Wess-Zumino-Witten (WZW) topological framing term:

$$\Gamma_{WZW} = \frac{k}{24\pi^2} \int_{\mathcal{M}_5} d^5x \epsilon^{ABCDE} \mathrm{Tr} \left(U^{-1}\partial_A U \ldots U^{-1}\partial_E U \right)$$

By applying the Finkelstein-Rubinstein theorem, the topological framing of the 3_1 Trefoil knot dictates that a full 2π spatial rotation forces the internal fluid ribbons to exchange. The WZW term mathematically mandates that this non-contractible rotation induces a strict -1 phase shift on the fluid

wavefunction:

$$\Psi \xrightarrow{2\pi \text{ rotation}} -\Psi$$

Any topological defect that acquires a -1 phase shift under a 2π rotation requires intrinsic angular momentum $J = 1/2$. Therefore, if two identical Trefoil knots (electrons) attempt to occupy the same quantum state, their phase fields destructively interfere to zero ($\Psi(r_1, r_2) = -\Psi(r_2, r_1)$). The Pauli Exclusion Principle is not an abstract quantum rule; it is derived within the stated model assumptions from the rotational topology of framed fluid knots.

2.4 Emergent Gravity and the Abram-Paschen Coupling

Varying the complete Master Action with respect to the bulk metric yields the stress-energy tensor T_{AB} of the vacuum. In the long-wavelength limit, the localized kinetic bending energy of topological defects behaves as an effective mass-energy source, mapping fluid pressure gradients into Ricci curvature ($R_{\mu\nu}$):

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}^{eff} = \frac{8\pi G_{eff}}{c_s^4} \left(T_{\mu\nu}^{(ideal)} + \Pi_{\mu\nu} \right)$$

where the viscoelastic stress tensor ($\Pi_{\mu\nu}$) enters the constitutive propagation model in the high-Reynolds, weak-dissipation limit.

Furthermore, the Abram-Paschen coupling is treated as a controlled macroscopic analogy: plasma-current routing responds to imposed acoustic-pressure gradients (∇P), providing a laboratory proxy for how charge transport may respond to substrate-pressure structure. This analogy constrains mechanism class, but it is not itself a proof of full quantum-gravitational closure.

Equation (2.1) serves as the generating action for the downstream limit reductions, so bulk, Chern-Simons, WZW, and boundary sectors remain traceable to promoted equation families.

By the same logic, spinorial and topological claims rooted in Eq. (2.3) are admissible only with circulation/framing closure in Eqs. (11.3.1) and (11.3.2), and with tensor-channel consistency enforced by Eqs. (11.11.1) and (11.11.2).

Part II

Galactic & Planetary Dynamics

3

Stability and the Landau Switch

3.1 The Thermodynamic Boundary of the Vacuum

A historic critique of treating the vacuum as a physical medium is the problem of planetary stability: *If spacetime is a fluid, why hasn't Earth lost its orbital energy to viscous drag and spiraled into the Sun?*

In Geotemporal Hydrodynamics (GTH), this is not an anomaly; it is a strict, derived consequence of macroscopic quantum mechanics. The substrate is a true superfluid. It possesses strictly zero viscosity as long as an object moves through it slower than the **Landau Critical Velocity** (v_{crit})—the threshold above which excitations (phonons/vortices) can be generated at no net energy cost.

For a system of characteristic scale R interacting with a substrate of healing length ξ, the finite-size Landau critical velocity is operationally defined as:

$$v_{crit}(R) \approx \frac{\hbar}{m_\psi R} \ln\left(\frac{R}{\xi}\right)$$

This equation dictates a profound scale-dependent regime separation. Because v_{crit} scales inversely with the interaction radius ($v_{crit} \propto R^{-1}$), the exact same fluid vacuum behaves as a frictionless, laminar void on small scales, yet transitions into a highly viscous, turbulent medium on massive scales.

3.2 The GTH Reynolds Parameter

To mathematically encode this regime switch, we define the GTH Reynolds-like parameter, comparing the characteristic velocity of a baryonic system (v) to the local Landau limit:

$$Re_{GTH}(R) \equiv \frac{v}{v_{crit}(R)}$$

- **The Subcritical Regime ($Re_{GTH} \ll 1$):** At the scale of the Solar System ($R \sim 1$ AU, $v \sim 30$ km/s), the Landau critical velocity is astronomically high ($v_{crit} \gg c$). The system is infinitely subcritical. The vacuum responds as a perfect, inviscid superfluid. Planetary orbits experience zero drag, preserving the precise Keplerian mechanics and local Lorentz invariance observed in standard General Relativity.

- **The Supercritical Regime ($Re_{GTH} > 1$):** At the scale of a spiral galaxy ($R \sim 10$ kpc, $v \sim 200$ km/s), the massive interaction radius forces the stability threshold to collapse. The rotation of the galaxy exceeds the Landau limit. The vacuum undergoes a thermodynamic phase shift—the "Landau Switch"—transitioning from a frictionless superfluid into a turbulent, wake-generating Navier-Stokes fluid.

This strict scale separation enforces cross-scale consistency: the same constitutive parameter tuple can produce galactic-scale wakes while suppressing measurable Solar-System drag. The mechanism is thresholded transport: below critical velocity, dissipation channels are closed and orbital motion remains effectively conservative; above threshold, nonlinear wake loading becomes active and contributes additional effective support. The scientific requirement is therefore not rhetorical consistency

but feasibility of one shared manifold across datasets. That feasibility is tested through joint inequalities linking Solar drag limits, galactic rotation residuals, and ringdown damping constraints.

The Landau threshold in Eq. (3.1) is carried forward as the scale-separation engine in Eq. (11.5.1) and in the feasibility inequalities Eqs. (11.6.3)–(11.6.3), so one regime switch governs Solar, galactic, and ringdown sectors alike.

Accordingly, the Reynolds criterion in Eq. (3.2) becomes a hard feasibility gate; tuples that fail joint drag, rotation, and ringdown constraints are removed from the admissible manifold.

4

Vorticity as Mass

4.1 The Illusion of the Dark Halo

Astrophysical observations universally demonstrate that the outer edges of spiral galaxies rotate significantly faster than their luminous baryonic mass permits under standard Newtonian dynamics. The orthodox ΛCDM model patches this discrepancy by hypothesizing a "Dark Matter Halo"—a massive, invisible sphere of collisionless particles. Despite decades of advanced direct-detection efforts, no particulate Dark Matter has ever been observed.

GTH abandons the particulate Dark Matter hypothesis entirely. The "missing mass" is not a new fundamental particle; it is the **Hydrodynamic Added Mass** of the galaxy's turbulent wake.

4.2 The Inverse Energy Cascade

A historic barrier to unifying Quantum and macroscopic turbulence pictures is the assumption that only forward cascades matter in relevant astrophysical regimes. The present framework instead uses an inverse-transfer channel to connect microscale strain injections to mesoscale and macroscale organized wake structure. This matters because flat rotation support requires coherent large-scale flow contributions, not merely local dissipation. The inverse-transfer claim is therefore central and must be confronted with resolved velocity-field diagnostics and alignment statistics in real survey data.

A historic barrier to connecting quantum and cosmological fluid dynamics has been the Richardson forward-cascade paradigm, which assumes turbulent kinetic energy transfers only from large scales down to dissipative microscale modes. Under that assumption, microscopic quantum vortices would not contribute to macroscopic galactic rotation support.

However, recent high-resolution computational fluid-dynamics analyses report inverse-transfer behavior in complex 3D strain fields. These analyses indicate that microscopic vortices can self-organize, synchronize vorticity, and transfer kinetic energy *upward* in scale, driving increasingly massive macroscopic flow structures.

This Inverse Energy Cascade is the exact fluid-mechanical engine of GTH cosmology.

1. **The Micro-State:** Individual fundamental particles (GeoKnots) operate in the subcritical regime, generating minute, localized strain fields ($\nabla\Psi$) in the substrate.

2. **The Synchronization:** When 10^{68} of these microscopic quantum knots aggregate into a rotating galactic disk, their individual strain fields physically overlap.

3. **The Macro-State:** Following this inverse-transfer mechanism, these microscopic vortices self-organize. Their combined localized strains cascade upward, generating a massive, unified macroscopic acoustic pressure gradient (∇P) that we observe as spacetime curvature.

4.3 The Reduced-Form Rotation Curve

Because the rotating galaxy operates in the supercritical regime ($Re_{GTH} > 1$), the viscous shear stress of the substrate transfers angular momentum outward, physically entraining a massive volume of the vacuum fluid.

The inertia of this entrained fluid vortex—the galactic wake—supplies the exact inward support traditionally attributed to Dark Matter. The corresponding rotation-curve velocity $v(r)$ is rigorously modeled by balancing the baryonic gravitational potential with the macroscopic wake tension:

$$v^2(r) = \frac{GM_b(r)}{r} + \beta r$$

The βr term is the direct mathematical manifestation of the Inverse Energy Cascade. It replaces the collisionless Dark Matter halo with a calculable, Navier-Stokes fluid wake coefficient (β, with units $\mathrm{m} \cdot \mathrm{s}^{-2}$).

4.4 Anisotropy and Wake-Shock Dynamics

Static, particulate Dark Matter models struggle to explain recent Gaia space observatory data, which reveals the Milky Way's "halo" is tilted, oblong, and asymmetric.

In GTH, this is a highly predictable, dynamic fluid response. To allow for non-axisymmetric wake geometry (e.g., satellite-induced tilts), the rotation curve incorporates a minimal phenomenological extension:

$$v^2(r, \vartheta) = \frac{GM_b(r)}{r} + \beta r \left[1 + \epsilon \cos(2(\vartheta - \phi_{LMC}))\right]$$

Here ϑ denotes azimuthal angle in the galactic plane. The parameters ϵ (anisotropy amplitude) and ϕ_{LMC} (orientation) physically model the hydrodynamic bow shock generated by the passage of the Large Magellanic Cloud (LMC). The LMC induces a transient shear strain that physically warps the fluid flow of the Milky Way's vacuum vortex, leaving a morphological scar (wake anisotropy) that is mathematically impossible to replicate with a static cloud of weakly interacting massive particles (WIMPs).

The wake-support closure in Eq. (4.3) is fitted jointly with the acceleration representation in Eq. (11.5.2) and with morphology diagnostics, so rotation-curve agreement is constrained by warp-phase and lensing-flow structure rather than amplitude matching alone.

In the same vein, the anisotropy extension in Eq. (4.4) is evaluated directly against residual-topology and temporal-coherence diagnostics in the observational workflow.

5

Macroscopic Isomorphisms

5.1 The Scale Invariance of Fluid Dynamics

GTH adopts conditional scale transport, not unrestricted scale invariance. The same hydrodynamic equations are expected only within a controlled asymptotic window where continuum closure, weak-compressibility ordering, and constitutive causality remain valid. Outside that window (quantum anomalies, chiral sectors, and strongly non-perturbative curvature), extra operators must be added explicitly rather than assumed to emerge automatically.

Accordingly, acoustic plasma guiding, OWPT, and RF harvesting are treated as mechanism-class analogs that calibrate transport intuition and similarity numbers; they do not, by themselves, prove full microphysical equivalence.

5.2 Acoustic Channels and Vacuum Gravity

In standard formulations, gravity and electromagnetism are modeled through distinct sectors. GTH treats selected transport responses as hydrodynamically coupled at the effective-medium level. Laboratory demonstrations of ultrasonic plasma guidance motivate this analogy for controlled transport studies.

By applying a high-intensity ultrasonic standing wave, an acoustic radiation force is generated that physically displaces the fluid medium, pushing heated, low-density gas toward acoustic antinodes. According to Paschen's Law, the electrical breakdown voltage V_b of a gas is a function of its density-dependent pressure p and the gap distance d:

$$V_b = \frac{B(pd)}{\ln(A(pd)) - \ln\left(\ln(1 + \frac{1}{\gamma_{se}})\right)}$$

Here A and B are the Paschen gas constants for the selected medium and γ_{se} is the effective secondary-emission coefficient at the electrode interface.

Because the acoustic field dynamically lowers the gas density at the antinodes, it locally minimizes the threshold required for electrical breakdown. The plasma propagates strictly along this low-density acoustic gradient.

The GTH Isomorphism: This is the macroscopic analog pattern of gravity–electromagnetic coupling. In GTH, a massive body acts as a volumetric sink, displacing the substrate and generating a continuous acoustic pressure gradient. The gravitational field is fundamentally an Acoustic Shear Wave in the vacuum. Electrons (charged Trefoil knots) are hydrodynamically forced to flow down this density gradient, exactly as the plasma in the Helsinki experiment is guided by ultrasonic antinodes.

5.3 Topological Funnels and Momentum Transfer

Optical Wireless Power Transfer (OWPT) and RF energy harvesting validate the topological momentum transfer of GTH. In RF harvesting, the conversion efficiency η_{RF} relies on rectifying diodes:

$$\eta_{RF} = \frac{V_{DC}^2/R_L}{P_{RF}}$$

In standard physics, photons propagate through an empty void. In GTH, ambient RF waves (P_{RF}) are macroscopic, low-frequency helicity ripples in the superfluid. The rectenna acts as a Topological Funnel. It absorbs the angular momentum of these open, unknotted fluid waves and utilizes that kinetic energy to physically bias the motion of knotted topological defects (electrons) through the lattice structure of the circuitry, generating V_{DC}.

5.4 The Viscous Wall: Interaction in a Wake-Supporting Medium

When a satellite galaxy moves through the host environment in a supercritical regime ($Re_{GTH} > 1$), standard collisionless dynamical friction (Chandrasekhar drag[6]) fails to accurately model the rapid orbital decay.

In the GTH wake medium, resistive coupling is governed by a turbulence-driven quadratic drag law:

$$F_{drag} = -\frac{1}{2} C_d \rho_0 A v^2 \hat{v}$$

The mechanical power dissipated by this quadratic drag scales with v^3:

$$\frac{dE}{dt} = F_{drag} \cdot v = -\frac{1}{2} C_d \rho_0 A v^3$$

This v^3 scaling dictates massive pericenter dominance: short, high-speed segments of the orbit contribute disproportionately to the cumulative energy loss. This is designed to reproduce the rapid orbital inspiral observed in close galactic mergers without relying on fine-tuned dark matter halo geometries.

In this chapter, the Abram-Paschen analogy and companion transport relations are treated as calibrated mesoscale discriminants with explicit failure routing in the falsifiability matrix.

Accordingly, Eqs. (5.2) and (5.3) are interpreted through similarity-controlled translation rather than one-step identity claims, while Eq. (5.4) enters the shared feasibility gate used in cross-channel calibration.

6

Viscoelastic Shear Hysteresis

6.1 Morphological Memory of the Vacuum

Standard astrophysical models assume that the morphology of a galaxy (warps, lopsidedness, stream asymmetries) is purely a function of its instantaneous gravitational potential. However, persistent, large-scale warps in outer stellar disks are difficult to explain as free kinematic tilts, because differential rotation should wind up the line of nodes over timescales far shorter than the galaxy's age.

In Geotemporal Hydrodynamics, the galactic disk is a boundary condition imposed on the surrounding medium. Because the substrate is a 5D viscoelastic fluid, it possesses thermodynamic memory. Morphological warps are interpreted as delayed-response structures rather than instantaneous equilibria, with phase lag controlled by relaxation scales and forcing history. This converts warp morphology into an inference channel: lag amplitude and orientation become estimators of constitutive memory parameters rather than purely geometric nuisance features.

6.2 The Israel-Stewart Relaxation Equation

To mathematically formalize this medium memory, we define the viscoelastic response of the substrate using the relativistic Israel-Stewart theory of dissipative fluid dynamics.

The Viscoelastic Stress Tensor ($\Pi_{\mu\nu}$) of the galactic wake does not respond instantaneously to changes in baryonic forcing. The delayed response is represented with a causal relaxation equation that separates source forcing from stress equilibration timescales:

$$\tau \Delta^{\alpha}_{\mu} \Delta^{\beta}_{\nu} \dot{\Pi}_{\alpha\beta} + \Pi_{\mu\nu} = -2\eta\sigma_{\mu\nu} - \zeta\Theta\Delta_{\mu\nu}$$

Where τ is the characteristic **Relaxation Time** of the fluid vacuum, η and ζ are the shear and bulk viscosities of the supercritical substrate, Θ is the expansion scalar, and $\sigma_{\mu\nu}$ is the shear strain induced by rotating galactic mass.

6.3 Predictions and Falsifiability

The τ parameter explicitly proves that the vacuum experiences **Shear Hysteresis**. If the relaxation time τ is long compared to the interaction timescale of a passing satellite galaxy (e.g., the LMC), the shear stress $\Pi_{\mu\nu}$ remains embedded in the medium long after the perturber has passed.

This generates ironclad, falsifiable morphological predictions. The phase of a galactic warp must correlate not just with the current position of massive perturbers, but with their historical trajectory through the viscous fluid. Furthermore, systems that operate strictly in the subcritical regime ($Re_{GTH} \ll 1$), where viscous terms functionally vanish, will not display these medium-driven morphological signatures. Vacuum memory is an exclusive property of the supercritical regime.

The causal memory model in Eq. (6.2) is linked directly to the relativistic control layer through Eqs. (11.6.1) and (11.6.1), preventing inconsistent viscosity assignments between galactic and ringdown regimes.

As a result, morphology-memory claims are judged by explicit delay and ringdown-consistency diagnostics in the observational analysis workflow.

Part III

Cosmology & Micro-Physics

7

The Big Freeze

7.1 The Top-Down Cosmological Paradigm

The standard cosmological narrative assumes the universe began from an infinitely dense, zero-dimensional singularity, with structure accreting slowly over billions of years from microscopic quantum fluctuations. This bottom-up accretion model is increasingly contradicted by James Webb Space Telescope (JWST) observations, which reveal supermassive, fully formed galaxies existing mere hundreds of millions of years after the Big Bang.

GTH resolves this crisis through a "Top-Down" hydrodynamic cosmology. The universe did not begin as a singularity; it began as a highly energetic, uniform thermal gas. The cosmological structure we observe today is the direct result of a massive, universe-wide thermodynamic phase transition.

7.2 The Electroweak Condensation

As the primordial universe expanded, the thermal energy of the environment dropped. When the ambient temperature crossed the critical Electroweak symmetry-breaking threshold ($T_{crit} \approx 246$ GeV), the scalar field underwent a violent Bose-Einstein Condensation. The gas rapidly "froze" into the macroscopic, coherent superfluid substrate state, acquiring its non-zero vacuum expectation value.

7.3 Defect Formation via the Kibble-Zurek Mechanism

Because the universe was expanding at a finite rate, this phase transition could not coordinate globally. According to the **Kibble-Zurek mechanism**[7, 8], causal domains froze out independently. The correlation length ξ and Kibble–Zurek relaxation time τ_{KZ} diverged near the critical point:

$$\xi(T) = \xi_0|\epsilon|^{-\nu}, \quad \tau_{KZ}(T) = \tau_{KZ,0}|\epsilon|^{-\mu}$$

At the boundaries where these mismatched, causally disconnected fluid domains intersected, the phase of the substrate order parameter was mathematically forced to singularity ($|\Psi|^2 \to 0$). This rapid freezing shattered the smooth vacuum, spawning a massive, universe-wide network of 1D topological defects (quantized cosmic vortex strings).

The initial defect density n is rigorously derived from the quench rate t_Q:

$$n \sim \frac{1}{\xi_{KZ}^3} \sim \left(\frac{\tau_{KZ,0}}{t_Q}\right)^{\frac{3\nu}{1+\mu}}$$

These primordial, chaotic vortex strings were the foundational raw material of all baryonic matter, providing massive, immediate density perturbations that seeded the early universe. The supermassive early galaxies observed by JWST are the direct, predicted consequence of the Kibble-Zurek defect network generated during the Electroweak Big Freeze.

The Kibble–Zurek relations in Eqs. (7.3) and (7.3) follow the same notation-lock discipline used elsewhere, preventing uncontrolled drift between early-universe and late-time sectors.

Predictions from this channel are promoted only after successful disjoint-set transfer tests, with full uncertainty routing carried through covariance propagation.

8

The Topological Standard Model

8.1 The Eradication of the Point Particle

Geotemporal Hydrodynamics (GTH) explicitly abandons the point-particle abstraction. In a superfluid vacuum, localized energy cannot collapse into a zero-dimensional point; it physically must form a **Quantized Topological Defect**—a stable vortex knot. The Standard Model of particle physics is a catalog of the lowest-energy topological geometries available to a 3D fluid.

To analyze particle stability, GTH evaluates the energy barrier for a quantum phase slip in the substrate. The probability P of spontaneous topological decay via thermal fluctuation is exponentially suppressed by the Boltzmann factor. At sub-atomic scales where the fluid velocity remains subcritical ($v \ll v_{crit}$), the effective acoustic temperature T_{eff} approaches zero. Reconnection and unspooling channels are therefore strongly suppressed in this regime.

8.2 Circulation, Helicity, and Emergent Charge

Internal quantum numbers in GTH are derived strictly from the classical ideal fluid invariants.

1. **Spin as Quantized Circulation:** The circulation around any closed vortex filament must be an integer multiple of the quantum of circulation κ. Internal quantum spin is the literal, quantized hydrodynamic circulation of the fluid around the defect core: $\Gamma = \oint \mathbf{v}_s \cdot d\mathbf{l} = n\kappa$.

2. **Electric Charge as Topological Writhe:** Utilizing the Călugăreanu-White-Fuller theorem ($L_k = T_w + W_r$), the projection of a knot's 3D topology onto the $U(1)$ electromagnetic subspace yields the charge operator. Charge Q is rigorously defined by the knot's internal core rotation (r) and spatial writhe (w):
$$Q = -\frac{e}{6}(w + r + 1)$$

8.3 The Lepton Class (Infinitely Subcritical Rings)

The Lepton class (e.g., Electrons) represents fully closed, self-sustaining vortex loops. The **Electron** is analytically defined as the foundational Left-Chiral Trefoil knot (3_1). Because the vortex line is topologically closed, there are no exposed boundaries to bleed pressure into the bulk substrate. Consequently, the localized fluid dynamics operate in a deeply subcritical regime ($Re_{GTH} \ll 1$).

In this limit, macroscopic viscous coupling is strongly suppressed. The electron therefore propagates with negligible drag and wake within the model approximation. This provides a hydrodynamic explanation for why leptons do not require the same confinement mechanism as quark sectors.

8.4 The Quark Class and Color Confinement

Unlike Leptons, Quarks are mathematically defined as **Open Vortex Filaments** (fractional braids). Because the vortex string is severed, the phase integral accumulates a fractional deficit, forcing the

internal framing to twist ($r = -2$) and generating the signature fractional charges of quarks via the Călugăreanu theorem.

The Proof of Color Confinement: In a superfluid substrate, an open vortex line cannot terminate in the bulk fluid. If an open string exists, the Dirichlet boundary conditions at the severed ends create a singularity where the fluid gradient energy diverges to infinity. Nature physically prohibits infinite energy gradients. To prevent this divergent energy bleed, Dirichlet boundary limits strictly mandate that the open ends of a Quark must violently snap onto the severed ends of *other* fractional vortex filaments to achieve mathematical closure. Quarks cannot exist in isolation.

8.5 The Baryon Assembly and the Strong Force

To achieve total topological closure, exactly three fractional filaments (e.g., two Up quarks, one Down quark) must link together. The resulting composite geometry is a **Borromean Link**—the foundational structure of the Proton (uud).

GTH eliminates the need for virtual messenger particles. The "Strong Force" is the emergent classical property of **Vortex Line Tension**. When an external force attempts to pull two braided quark filaments apart, it stretches the vortex tube. The restoring force is simply the linear tension of the fluid attempting to minimize the total volume of the zero-density phase defect, yielding the linear confinement term (σr).

The topological particle classification is tethered to the quantization and persistence machinery in Eqs. (11.3.1)–(11.3.3), so spin, charge, and stability assignments are constrained by framed-defect transport rather than by analogy alone.

Accordingly, charge and confinement statements based on Eq. (2) must satisfy explicit derivation consistency and shared-parameter closure tests.

Part IV

Observational Verification

9

Computational Verification

9.1 The Cross-Scale Consistency Mandate

A central requirement of this framework is cross-scale consistency: the same governing equations should remain coherent from microscopic to cosmological regimes under declared limits. To test that requirement, an automated computational pipeline evaluates the substrate state across four physical regimes using an illustrative ultralight benchmark ($m_\psi \approx 1.2 \times 10^{-22}$ eV). In this manuscript revision, that benchmark is provisional and is not treated as a fixed gravity-normalization closure point.

9.2 The Computational Pipeline Analysis

I. The Quantum Scale (Bohr-Radius Baseline)
At the atomic level ($R \approx 5.29 \times 10^{-11}$ m, $v \approx 2.18 \times 10^6$ m/s), applying the Landau equation yields a very large critical velocity. The quantum regime is deeply subcritical ($Re_{GTH} \ll 1$), so vacuum-drag corrections remain below the model's practical resolution target in this sector.

II. The Solar System Scale (Keplerian Baseline)
Evaluating Earth's orbital parameters ($R \approx 1.5 \times 10^{11}$ m, $v \approx 30$ km/s), the critical velocity remains far above local flow speeds. The Solar System stays subcritical, so no macroscopic wakes are predicted; this is consistent with null local direct-detection signatures and with high-precision Newtonian/post-Newtonian orbital fits.

III. The Galactic Scale (SPARC Baseline[9])
Evaluating a typical spiral galaxy ($R \sim 10$ kpc, $v \sim 200$ km/s), the massive interaction radius forces the stability threshold to collapse. The characteristic velocity exceeds the critical limit ($v_{gal} > v_{crit}$). The system transitions into the supercritical regime ($Re_{GTH} > 1$). The Inverse Energy Cascade is triggered, generating the macroscopic entrained wake (βr) that mimics a Dark Matter halo.

IV. The Near-Horizon Scale (GW250114 Baseline)
For a massive binary black-hole merger ($R_{horizon} \approx 1.85 \times 10^5$ m and $v \approx 0.5c$), the compact interaction radius yields a very large v_{crit}. The response remains subcritical ($v \ll v_{crit}$), vortex nucleation is energetically suppressed, and the dynamics remain predominantly laminar in this approximation.

The four-regime computational narrative is executed as a reproducible inference program in which every benchmark maps to manifest-linked artifacts and deterministic replay checks.

Operationally, outputs are admissible only when cross-channel closure checks are satisfied and full run-record requirements are met.

10

Observational Verification: GW250114

10.1 The Ringdown Stress Test

A high-value empirical stress test for any fluid-vacuum model is gravitational-wave propagation. A standard concern is that a viscous medium could overdamp high-frequency ringdown structure relative to LIGO constraints.[10–12] Geotemporal Hydrodynamics incorporates this constraint and uses GW250114 as a consistency test for Kerr-like ringdown spectroscopy within measurement error.

10.2 Viscoelastic Immunity and the Israel-Stewart Resolution

In standard General Relativity, a merger is described geometrically. In GTH, a binary black-hole merger is modeled as a catastrophic acoustic-cavitation event.

As established in the computational pipeline, the near-horizon scale operates in a subcritical regime ($Re_{GTH} \ll 1$). In this regime the medium response is predominantly elastic, and propagation of the resulting acoustic-shear perturbations is modeled with the Israel–Stewart relaxation equation[13, 14]:

$$\tau \Delta^{\alpha}_{\mu} \Delta^{\beta}_{\nu} \dot{\Pi}_{\alpha\beta} + \Pi_{\mu\nu} = -2\eta\sigma_{\mu\nu}$$

During a merger, the localized shear rate $(1/\Delta t)$ can exceed the intrinsic relaxation timescale of the vacuum fluid. In the high-frequency limit $(\omega\tau \gg 1)$, effective viscous dissipation is suppressed, so the substrate response is modeled as elastic-dominant rather than viscosity-dominant.

10.3 Absolute Cross-Scale Coherence

The analytical outcome of the constitutive model is that ringdown-scale damping can be strongly suppressed while galactic-scale wake channels remain active in supercritical regimes. GW250114 is therefore compatible with the fluid-vacuum hypothesis under the stated parameter window, but compatibility alone is not uniqueness; discriminant predictions against alternative models remain essential. The framework's core claim is a single constitutive manifold that can jointly account for kpc-scale wake behavior and near-horizon ringdown fidelity without ad hoc sector-specific constants.

The ringdown relation in Eq. (10.2) is constrained by the relativistic dissipation window in Eqs. (11.6.3) and (11.6.3), with event-level diagnostics tied to the same constitutive tuple that governs galactic wakes.

Hence compatibility is accepted only after explicit envelope and rejection tests across ringdown and cross-channel residual structure; failures exclude the fit from the admissible manifold.

10.4 Program Work Packages

To keep development rigorous while remaining executable, the program is decomposed into six bounded work packages:

W1: Foundational Closure. Lock the action-to-equation derivation chain and enforce tensor/sign conventions before parameter fitting.

W2: Regime-Switch Dynamics. Couple Landau-threshold mechanics, wake transport, and morphology diagnostics under one shared inequality set.

W3: Analogy Discrimination. Convert mechanism analogies and memory claims into predeclared decision boundaries and lag-envelope falsifiers.

W4: Microphysics Closure. Bind topological particle sectors to quantization, framing, and consistency constraints.

W5: Computational Reproducibility. Require deterministic replay and equation-to-code traceability for every promoted result.

W6: Global Falsification Pass. Enforce ringdown envelopes jointly with cross-channel residual and admissibility gates.

A work package is accepted only if it passes hard-feasibility screening together with cross-channel rejection criteria under fixed, predeclared thresholds.

Part V

Derivation Program and Open Proof Tasks

This part supplies the formal derivations, scale-separation machinery, computational reproducibility practices, and explicit failure surfaces that carry the preceding narrative into testable closure.

11

Substrate Field Closure and Stability Proofs

11.1 Variational Backbone of the Substrate Sector

11.1.1 Functional Domain and Boundary Class

We formalize the substrate order parameter on a Sobolev class $\Psi \in H^1(\mathcal{M}_5)$ with finite action and finite helicity norm. The admissible domain is restricted by bounded boundary flux and finite circulation spectra. This choice is not cosmetic; it is the minimal regularity needed to keep both Gross-Pitaevskii energetics and topological invariants simultaneously well-defined.

In this setting, phase singularities correspond to codimension-two defects and are represented by measure-valued vorticity currents. The regular component of Ψ evolves under smooth variational flow, while the defect skeleton follows constrained transport generated by circulation quantization.

11.1.2 Euler-Lagrange System and Constraint Algebra

Starting from the covariant action, stationarity under $\delta\Psi^\dagger$ yields

$$D_A D^A \Psi + \mu^2 \Psi - \lambda |\Psi|^2 \Psi + \mathcal{J}_{top} = 0,$$

where $\mathcal{J}_{top}$ collects framed-topology backreaction terms. Variation with respect to the auxiliary gauge potentials closes the Chern-Simons-like helicity channel and imposes algebraic compatibility between circulation and writhe sectors.

The resulting system is a constrained hyperbolic-elliptic pair: hyperbolic in propagating acoustic modes, elliptic in static pressure closure. This split is exactly what allows low-frequency cosmological drift and high-frequency near-horizon elastic response to coexist.

11.1.3 Energy Coercivity and Lyapunov Monotonicity

Define the coarse-grained free-energy functional

$$\mathcal{F}[\Psi] = \int_{\mathcal{M}_5} \left(\frac{1}{2} |D\Psi|^2 + V(|\Psi|^2) \right) d^5X + \int_{\partial\mathcal{M}} \gamma(\Psi)\, d^4x.$$

For admissible data with bounded defect density, $\mathcal{F}$ is coercive modulo gauge orbit and framing class. Under dissipative coarse-graining, one obtains $\dot{\mathcal{F}} \leq 0$ outside driven boundaries, yielding a Lyapunov structure for metastable geoknot sectors.

This monotonicity is the analytic reason particle-like topological defects remain persistent over macroscopic timescales without being fundamental point objects.

11.2 Constructive Derivations for G_{eff} and m_e

11.2.1 Long-Wavelength Reduction and Effective Poisson Coupling

Linearize the substrate phase around a stationary background, writing $\Psi = \sqrt{\rho_0 + \delta\rho}\, e^{i(\theta_0+\varphi)}$ and keeping terms up to quadratic order in perturbations. After eliminating $\delta\rho$ with the continuity constraint, the reduced action becomes

$$S_\varphi = \int d^4x \left[\frac{\rho_0}{2c_s^2}(\partial_t\varphi)^2 - \frac{\rho_0}{2}(\nabla\varphi)^2 + g_\psi\, \varphi\, \rho_m \right],$$

where ρ_m is the coarse-grained defect mass density and g_ψ is the substrate-defect coupling. In the static limit $\partial_t\varphi \to 0$, stationarity gives

$$\nabla^2\varphi = -\frac{g_\psi}{\rho_0}\rho_m.$$

Defining the effective potential as $\Phi \equiv -g_\psi\varphi$ and matching the Poisson form $\nabla^2\Phi = 4\pi G_{eff}\rho_m$ yields

$$G_{eff} = \frac{g_\psi^2}{4\pi\rho_0}.$$

Using the circulation-normalized coupling closure

$$g_\psi^2 = \frac{3\pi^2\rho_0\hbar c_s}{m_\psi^2},$$

one obtains

$$G_{eff} = \frac{3\pi\hbar c_s}{4m_\psi^2},$$

which is the executive-summary expression.

11.2.2 Line-Tension Closure Derivation of m_e

Take the knot closure ansatz

$$L_{knot} = \frac{h}{m_e c}, \qquad \mathcal{T}_{vacuum} = \frac{\rho_0 h^2}{4\pi m_\psi^2}\ln(\Lambda).$$

The closure-work model sets rest energy equal to line-tension work:

$$m_e c^2 = \mathcal{T}_{vacuum} L_{knot}.$$

Substituting the two expressions above gives

$$m_e c^2 = \left(\frac{\rho_0 h^2}{4\pi m_\psi^2}\ln(\Lambda)\right)\left(\frac{h}{m_e c}\right) \Rightarrow m_e^2 = \frac{\rho_0 h^3}{4\pi c^3 m_\psi^2}\ln(\Lambda),$$

and therefore

$$m_e = \sqrt{\frac{\rho_0 h^3}{4\pi c^3 m_\psi^2}\ln(\Lambda)}.$$

11.2.3 Identifiable Combinations and Non-Uniqueness

The data do not identify all primitive parameters independently; they identify combinations

$$\Xi_1 \equiv \frac{c_s}{m_\psi^2}, \qquad \Xi_2 \equiv \frac{\rho_0\ln(\Lambda)}{m_\psi^2},$$

through $G_{eff} \propto \Xi_1$ and $m_e^2 \propto \Xi_2$. Hence current closure is a constrained reparameterization unless additional observables break these degeneracies.

11.3 Topological Quantization and Knot Persistence

11.3.1 Circulation Spectrum and Integer Classes

Quantized circulation obeys

$$\Gamma_n = \oint_C \mathbf{v}_s \cdot d\mathbf{l} = n\kappa, \qquad \kappa = \frac{h}{m_\psi}, \quad n \in \mathbb{Z}.$$

The integer class labels disconnected homotopy sectors of admissible maps from loop contours into substrate phase space. Continuous evolution cannot change n without crossing a singular reconnection event.

11.3.2 Framed Trefoil Sector and Spinorial Sign Rule

For framed trefoil defects, the framing bundle contributes a nontrivial lift under 2π rotation. The Finkelstein-Rubinstein sign is encoded as a topological phase holonomy:

$$\mathrm{Hol}_{2\pi}(\mathcal{L}_{frame}) = -1.$$

Hence wavefunction transport around the rotational loop returns with odd sign, enforcing half-integer spin statistics in the defect sector.

11.3.3 Reconnection Threshold and Exponential Suppression

A phase-slip pathway exists only if the local energetic barrier is crossed:

$$P_{slip} \sim \exp\left(-\frac{\Delta E_{core}}{k_B T_{eff}}\right).$$

In deeply subcritical regimes where $T_{eff} \to 0$, reconnection probabilities are exponentially negligible, producing effective topological immortality for low-lying leptonic knots.

11.4 Mass-Energy Reconstruction from Line Tension

11.4.1 Geometric Closure at Compton Resonance

The electron-class defect is modeled by a closed arc-length resonance condition

$$L_{knot} = \frac{h}{m_e c}.$$

This links rest mass to geometric phase closure rather than to an independent particle postulate.

11.4.2 Line-Tension Work and Rest-Energy Equivalence

With vacuum line tension

$$\mathcal{T}_{vacuum} = \frac{\rho_0 h^2}{4\pi m_\psi^2} \ln(\Lambda),$$

closure work gives

$$m_e c^2 \sim \mathcal{T}_{vacuum} L_{knot},$$

which yields the derived electron mass relation already stated in the executive summary.

11.4.3 Perturbative Sensitivity and Error Channels

Logarithmic cutoff sensitivity enters only through $\ln(\Lambda)$, making the mass sector slowly varying under ultraviolet uncertainty. Dominant uncertainty channels are therefore substrate density calibration and scalar mass prior m_ψ, not catastrophic UV divergence.

The derivation program now extends from defect-scale closure to astrophysical transport closure: the same constitutive tuple must simultaneously account for local drag suppression, galactic wake support, and morphology-level discriminants without retuning.

11.5 Scale Separation, Drag Regimes, and Halo Morphology

With defect-scale closure established, we now formalize the radius-dependent switch that lets one constitutive medium appear drag-free in compact systems yet wake-supporting at galactic scales.

11.5.1 Landau Threshold Mechanics Across Radii

Finite-Size Critical Velocity Model

The operational threshold

$$v_{crit}(R) \approx \frac{\hbar}{m_\psi R} \ln\left(\frac{R}{\xi}\right)$$

implies monotone decline with increasing interaction radius. The same medium can thus be locally inviscid and globally wake-generating, depending only on scale.

Subcritical Planetary Sector

For AU-scale systems, $v/v_{crit} \ll 1$. Excitation channels are frozen, drag terms are asymptotically suppressed, and orbital mechanics remain effectively Keplerian within observational precision.

Supercritical Galactic Sector

For kpc-scale rotation, threshold crossing $v/v_{crit} > 1$ opens turbulent transfer channels. The medium supports entrained wakes and effective halo-like support without particulate dark-matter assumptions.

11.5.2 Inverse Cascade Transfer and Effective Halo Terms

Two-Stage Cascade Picture

Microscopic strain defects seed mesoscopic vortical structures. Nonlinear coupling then drives an inverse transfer of kinetic organization toward macroscopic coherent flow envelopes.

Phenomenological Rotation-Curve Closure

The effective radial acceleration model can be represented as

$$a_{eff}(r) = \frac{GM_b(r)}{r^2} + \beta,$$

with β interpreted as acceleration-level substrate wake loading rather than unseen collisionless mass.

Morphology Predictions and Phase Memory

Because the wake is a dynamical field, warped disks and asymmetric tails inherit trajectory memory. The framework predicts morphology-lag correlations between present baryonic geometry and prior perturber paths.

11.5.3 Dwarf, Cluster, and Transitional Windows

Dwarf-Galaxy Edge Cases

Low-velocity dwarfs can straddle near-critical thresholds where small environmental changes toggle partial wake formation. This predicts heterogeneous halo signatures for similar baryonic masses.

Group and Cluster Coupling

At cluster scales, overlapping wakes create collective transport channels. Effective drag and pressure support become networked rather than single-host quantities.

Observational Discriminants

Key discriminants include wake alignment fields, shear-lag statistics, and cross-correlation of halo asymmetry with reconstructed flow history. These are testable in multi-survey kinematic maps.

To remain non-contradictory across scales, the same closure manifold must also preserve compact-object ringdown structure in high-frequency regimes.

11.6 Relativistic Dissipation Control and Ringdown Fidelity

At relativistic frequencies, the same constitutive sector must satisfy causal dissipation bounds while preserving observable ringdown fidelity.

11.6.1 Israel-Stewart Sector in Substrate Variables

Constitutive Equation and Causal Closure

We adopt the relaxation form

$$\tau \Delta^{\alpha}_{\mu} \Delta^{\beta}_{\nu} \dot{\Pi}_{\alpha\beta} + \Pi_{\mu\nu} = -2\eta\sigma_{\mu\nu}$$

for shear stresses in the effective medium. This avoids acausal diffusion pathologies of first-order relativistic hydrodynamics.

Frequency-Domain Interpretation

In Fourier space, $\Pi(\omega)$ behaves with transfer factor $(1 + i\omega\tau)^{-1}$. For $\omega\tau \gg 1$, viscous leakage is parametrically suppressed and elastic response dominates.

Ringdown Compatibility Condition

GW ringdown preservation requires damping rates well below current detectability thresholds. The substrate condition is

$$\frac{\eta}{1 + (\omega\tau)^2} \ll \eta_{crit},$$

which is naturally satisfied in near-horizon high-frequency sectors.

11.6.2 Near-Horizon Regime Matching

Characteristic Scale Separation

Merger-horizon scales are compact enough that local critical velocity remains high relative to induced flow perturbations. This preserves subcritical behavior despite relativistic orbital speeds.

Quasinormal Mode Integrity

Mode frequencies track Kerr-like spectra when dissipation is subleading to elastic restitution. Within the admissible window, the model remains consistent with observed ringdown sharpness while retaining fluid ontology.

Post-Merger Relaxation Tail

Late-time tails correspond to slow relaxation of residual anisotropic stresses rather than to large viscous decay. This yields subtle but testable phase-shift corrections in high-SNR events.

11.6.3 Cross-Scale Non-Contradiction Theorem

Quantitative Feasibility Constraints

Define the effective dissipative coefficient

$$\eta_{eff}(\omega, v, R) = \frac{\eta}{1 + (\omega\tau)^2} + \eta_{turb}\, H\left(\frac{v}{v_{crit}(R)} - 1\right),$$

with H the Heaviside switch. A single parameter tuple is admissible only if

$$\eta_{eff}(\omega_{orb}, v_{orb}, R_{AU}) \leq \eta_{\odot}^{max},$$

$$\eta_{eff}(\omega_{rd}, v_{rd}, R_{BH}) \leq \eta_{GW}^{max},$$

$$\beta(\rho_0, v_{gal}) \equiv C_w \rho_0 v_{gal}^2 \geq \beta_{min}.$$

These are the Solar-drag, ringdown, and galactic-wake constraints in directly testable form, with C_w carrying units of $\mathrm{m}^2\,\mathrm{kg}^{-1}$ so that β has acceleration units.

Constructive Non-Empty Window Criterion

A sufficient feasibility set is

$$v_{orb} < v_{crit}(R_{AU}), \qquad v_{gal} > v_{crit}(R_{gal}), \qquad \tau \geq \frac{1}{\omega_{rd}}\sqrt{\frac{\eta}{\eta_{GW}^{max}} - 1},$$

$$\rho_0 \geq \frac{\beta_{min}}{C_w v_{gal}^2}.$$

If these inequalities are simultaneously satisfied, the admissible manifold is provably non-empty.

No-Drag / Strong-Wake Compatibility

A central closure claim of the framework is that subcritical compact systems and supercritical galactic systems are not contradictory states, but distinct points on the same scale-dependent constitutive manifold.

Parameter Window Feasibility

There exists nonempty parameter volume in $(m_\psi, \xi, \tau, \eta)$ such that Solar-system drag is negligible, ringdown remains crisp, and galactic wakes are strong.

Falsification Surface

The theory is falsified if any single parameter set fails to satisfy all three constraints simultaneously under calibrated observational priors.
Having established variational, topological, and cross-scale constitutive closure, the manuscript now turns to the computational and empirical infrastructure required to make those claims reproducible, transparent, and genuinely falsifiable.

11.7 Numerical Pipeline and Reproducibility Framework

These constitutive conditions are then translated into a reproducible computational workflow with deterministic validation gates.

11.7.1 Dimensionless Recasting and Solver Topology

Nondimensional Groups

Define characteristic scales (L_0, V_0, ρ_0) and derive dimensionless controls $\Pi_1 = Re_{GTH}$, $\Pi_2 = \omega\tau$, and $\Pi_3 = \xi/L_0$. This reduces apparent parameter freedom and exposes universal regime maps.

Mesh Strategy and Stability Criteria

Adaptive mesh refinement is centered on defect cores and shear interfaces. Time stepping satisfies CFL-like criteria for acoustic modes and separate relaxation stability bounds for viscoelastic channels.

Verification Tests

Code-level verification requires manufactured-solution checks, circulation conservation tests, and convergence under mesh refinement in both laminar and supercritical benchmark flows.

11.7.2 Data Assimilation from Astrophysical Catalogs

Rotation-Curve Inference Loop

Given baryonic profile priors, infer wake parameter fields by minimizing predeclared residual objectives between predicted and observed circular velocities across radius bins.

Ringdown Constraint Inference

Map posterior samples of (η, τ) against event-level ringdown damping limits. Reject samples violating quasinormal fidelity thresholds.

Joint Posterior Construction

Construct a joint posterior coupling galaxy, lensing-morphology, and GW channels. This enforces cross-scale self-consistency at inference time.

11.7.3 Reproducible Artifact Protocol

Configuration Manifests

Every simulation run must store full manifests: parameter values, random seeds, mesh hashes, solver versions, and boundary condition signatures.

Deterministic Replay

Replay scripts must reproduce numerics bitwise where hardware permits, otherwise within strict floating-point envelopes documented per benchmark.

Result Trace Tables

Each figure and equation-linked result must be accompanied by a trace table linking textual claims to run identifiers and statistical confidence summaries.

11.8 Empirical Bridge: Plasma-Guiding Isomorphisms

The empirical bridge is treated as a controlled transport analogy: sufficient to test mechanism structure, but not interpreted as constituent-level identity.

11.8.1 Abram-Paschen Mapping as a Control Analogy

Acoustic Gradient Forcing in Plasma Columns

Laboratory plasma columns subjected to controlled acoustic density gradients exhibit directed charged-flow steering. This furnishes a macroscopic analog for pressure-gradient guided transport.

Constitutive Correspondence Rules

A minimal dictionary maps laboratory observables to substrate variables: acoustic-pressure gradients to effective geodesic forcing, plasma-current shear to charge transport channels, and boundary compliance to medium response class.

Scaling Translation

Scale translation uses dimensionless similarity, not naive unit matching. Equivalent behavior is established through matched control numbers and response topology.

11.8.2 Experimental Design for Discrimination

Null-Hypothesis Construction

The null model assumes no directed coupling between imposed acoustic gradient and net charge-channel curvature after correcting for instrumental drift.

Signal Extraction Pipeline

Signal extraction applies phase-locked averaging and cross-spectral coherence to separate deterministic steering from stochastic turbulence.

Decision Boundaries

Detection thresholds are defined by pre-registered Bayesian factors and effect-size floors to avoid a posteriori overfitting.

11.8.3 Interpretive Limits and Scope

Where the Analogy Holds

The mapping is strongest for transport topology and forcing geometry, where shared hydrodynamic structure dominates model dependence.

Where the Analogy Breaks

The analogy does not claim identity of microscopic constituents; it claims structural equivalence of governing response classes.

Integration into the Unified Test Stack

Plasma-guiding analogs serve as mesoscale validation of the coupling logic, complementing astrophysical and gravitational-wave constraints.

11.9 Extended Falsifiability Matrix and Research Program

Falsifiability is then encoded as ranked test classes, explicit rejection boundaries, and campaign-level priorities.

11.9.1 Prediction Classes and Priority Ranking

Class-I Immediate Tests

Class-I tests use existing datasets: rotation-curve residual anisotropy, halo-wake lag metrics, and ringdown damping envelopes.

Class-II Targeted Campaigns

Class-II tests require new observational programs: high-cadence warp-history reconstructions and coordinated multi-band kinematic maps.

Class-III Long-Horizon Tests

Class-III tests include precision secular-drift baselines and ultra-high-SNR GW catalogs for subtle post-merger phase diagnostics.

11.9.2 Model Failure Modes

Parameter Incompatibility Failure

If no joint posterior region satisfies Solar-system, galactic, and GW constraints simultaneously, the framework fails globally.

Morphology-Causality Failure

If observed morphology-lag relations are absent or opposite to predicted wake-memory signatures, the cascade interpretation fails.

Transport-Analogy Failure

If controlled plasma-guiding experiments show no reproducible gradient-charge steering under matched similarity controls, the empirical bridge fails.

11.10 Technical Appendices in Narrative Form

The appendices keep narrative readability while preserving derivation traceability, notation discipline, and clear interpretability.

11.10.1 Core Derivation Sketches

Acoustic Metric Emergence Sketch

Linearizing around background density and phase yields a wave operator on perturbations equivalent to propagation on an effective metric $g_{\mu\nu}^{eff}(\rho, \mathbf{v}_s)$. Concretely, writing $\rho = \rho_0 + \delta\rho$ and $\theta = \theta_0 + \varphi$, the quadratic action in $(\delta\rho, \varphi)$ produces a hyperbolic equation with characteristic cone set by c_s and advective shift set by $\mathbf{v}_s$. After eliminating $\delta\rho$ through the continuity constraint, the principal symbol defines an acoustic metric of Unruh-type with lapse, shift, and spatial metric all determined by substrate background fields. This provides the technical bridge between microscopic phase fluctuations and emergent geodesic structure used throughout the manuscript.

Stress-Energy Decomposition

The effective stress tensor decomposes into ideal, anisotropic, and relaxation sectors. The ideal sector tracks barotropic pressure and kinetic flow; the anisotropic sector captures defect-induced shear alignment; and the relaxation sector introduces causal memory through Israel–Stewart-type terms. Each sector is traceable to explicit terms in the master action and constitutive closure, enabling equation-to-code traceability in the numerical pipeline. In practical inference, this decomposition is essential because different observables constrain different sectors: rotation curves constrain effective pressure loading, while ringdown data constrain relaxation and shear-memory channels.

Boundary Work and Dark-Energy Sign

Boundary work against positive surface tension contributes negative effective pressure in the expanding manifold picture, reproducing the sign structure required for accelerated expansion. The key book-keeping step is to separate bulk kinetic terms from boundary free-energy terms and then compute the equation-of-state contribution $w_{eff} = p_{eff}/\rho_{eff}$ under slow expansion. In this limit, boundary-work terms enter with opposite sign relative to bulk compression terms, producing an effective negative pressure channel without requiring a fundamental cosmological constant insertion by hand.

11.10.2 Notational Harmonization

Symbol Registry

A single registry maps all symbols to physical meaning and dimensional units to eliminate cross-chapter ambiguity. The registry should include: symbol name, tensor rank, unit dimension, governing equation references, and associated implementation identifiers in code. This prevents semantic drift where the same symbol acquires different operational meanings in different chapters. It also permits automated dimensional checks and manuscript-to-notebook validation.

Regime Labels

Regimes are tagged by $(Re_{GTH}, \omega\tau)$ quadrants to keep drag and dissipation language consistent across scales. The first coordinate controls subcritical versus supercritical transport; the second controls viscous-dominant versus elastic-dominant response. By assigning every dataset and simulation run to one quadrant, interpretation errors are reduced and cross-scale claims become traceable with explicit regime labels.

Operator Conventions

Covariant derivatives, projections, and overdot conventions are standardized so tensor equations remain syntactically and semantically aligned. Specifically, overdots denote flow-frame proper-time

derivatives, projected gradients use $\Delta_{\mu\nu}$, and all stress tensors are written with consistent index placement and metric signature. This standardization is required for reproducible symbolic manipulation and to avoid hidden sign errors in constitutive terms.

11.10.3 Reader Navigation Layer

How to Read for Physics First

Readers focused on physical claims can follow the executive summary, Landau-switch chapter, and GW verification chapter before deep formal sections. This path highlights the observational motivation, the regime-switch mechanism, and ringdown compatibility before engaging full functional analysis. It is intended for readers prioritizing phenomenology and falsifiable consequences over full formal derivation details.

How to Read for Mathematical Rigor

Readers focused on derivations should proceed from master action, topological closure, and viscoelastic constitutive sectors before empirical mapping. This order preserves dependency structure: action-level assumptions define Euler–Lagrange equations; topological constraints define admissible defect classes; constitutive closure defines transport and damping behavior. Empirical sections should then be read as consequence checks of that formal backbone.

How to Read for Falsifiability

Readers focused on testability should prioritize the falsifiability matrix, inference pipeline, and discriminant prediction classes. Operationally, this means starting from measurable inequalities, then examining posterior construction and robustness sweeps, and finally evaluating forecast discriminants against baseline models. This path supports direct assessment of whether the framework adds unique predictive value.
The following sections deepen tensor closure, observational synthesis, implementation practice, and quantitative exclusion logic so that the closure program remains coherent from equation level to pipeline level.

11.11 Tensorial Closure Details

Tensor-level identities and asymptotic ordering rules are collected here to maintain consistency across inference and simulation.

11.11.1 Projected Dynamics in 4+1 Splitting

Projector Identities

Define u^μ as the effective flow 4-velocity and $\Delta_{\mu\nu} = g^{eff}_{\mu\nu} + u_\mu u_\nu$ as the orthogonal projector. Repeated use of projector identities yields clean separation between compressive and shear channels in the constitutive system. In particular, contractions with u^μ isolate energy transport, while double projection isolates spatial stresses in the comoving frame. This decomposition is used to derive causal transport equations without mixing frame artifacts.

Shear-Bulk Decomposition

Write

$$\nabla_\mu u_\nu = \sigma_{\mu\nu} + \omega_{\mu\nu} + \frac{1}{3}\Theta\Delta_{\mu\nu} - u_\mu a_\nu,$$

with symmetric-tracefree shear $\sigma_{\mu\nu}$, vorticity $\omega_{\mu\nu}$, expansion Θ, and acceleration a_ν. This decomposition is used consistently for all ringdown and wake calculations.

Constraint Propagation

Gauge and topological constraints propagate if satisfied initially under the chosen evolution scheme. Numerically, this is monitored through residual norms of divergence and helicity constraints at each time step. A practical guardrail is to require bounded residual growth under mesh refinement; monotone growth indicates either an inconsistent discretization or inadequate constraint cleaning.

11.11.2 Curvature-Pressure Correspondence

Effective Einstein Mapping

At long wavelength, substrate stress maps to curvature through

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}^{eff} = \frac{8\pi G_{eff}}{c_s^4} \left(T_{\mu\nu}^{(ideal)} + \Pi_{\mu\nu} \right).$$

The relation is interpreted as constitutive emergence, not geometric fundamentalism.

Acoustic Geodesic Correction Terms

Subleading terms include gradients of density and phase curvature that perturb effective geodesic paths. These corrections are small in laminar domains and larger near strong shear transitions, where higher-order derivative terms in the effective action are amplified. Quantitatively, correction size is controlled by powers of ξ/L and local strain gradients, enabling controlled truncation in weak-shear regimes.

Regime Ordering

A consistent ordering in powers of (ξ/L) and Mach-like flow ratios keeps expansions controlled and prevents mixed-order inconsistency. Every retained term is assigned an asymptotic order, and discarded terms are explicitly tracked as remainder classes. This makes approximation error explicit and prevents accidental combination of incompatible truncation levels.

11.11.3 Helicity and Charge Sector Couplings

Writhe-Rotation Charge Encoding

Charge encoding follows topological invariants in framed defects. The operational relation to writhe and internal rotation remains fixed across chapters for symbol consistency, so that sign conventions and parity assignments do not drift between formal and observational sections. This is crucial for any claim connecting topology to measurable electromagnetic response.

Helicity Flux Balance

Helicity flux balance equations determine when defect bundles remain coherent versus when they undergo cascade-mediated redistribution. In practice, coherence persists when injected helicity exceeds dissipative leakage across the same scale band; redistribution dominates when nonlinear transfer terms exceed retention terms. This criterion directly informs wake morphology expectations.

Gauge-Compatible Regularization

Regularization is applied in a gauge-compatible manner so conserved quantities remain invariant under admissible transformations. Counterterms are introduced only in combinations that preserve the constraint algebra and do not alter topological class labels. This ensures that renormalized quantities remain physically comparable across scales and numerical resolutions.

11.12 Observational Synthesis Atlas

This atlas organizes cross-channel diagnostics for testing whether one shared constitutive tuple survives projection across scales.

11.12.1 Galaxy-Scale Diagnostics

Rotation Curve Families

The model classifies curves into subcritical, near-critical, and supercritical families. Each family has distinct residual morphology under baryonic subtraction: subcritical systems show near-Newtonian tapering, near-critical systems show transition shoulders, and supercritical systems show persistent outer flattening with asymmetric wake signatures. This taxonomy enables direct stratified fitting instead of forcing a single residual template on all galaxies.

Warp-Phase Memory Tracking

Within GTH, warp phase is expected to correlate with historical perturber passage through a delayed-response signature characteristic of viscous-memory media. The delay kernel depends on relaxation scales and local shear history, so the inferred lag should vary systematically with host mass and interaction geometry. This yields a concrete discriminant: a morphology–memory phase lag that is not naturally produced by purely collisionless, static-halo models.

Lensing-Flow Joint Features

When inverse-cascade entrainment dominates, weak-lensing elongation and kinematic shear should co-align with reconstructed wake axes. Misalignment statistics therefore provide a direct test of whether inferred effective mass is tied to flow structure or instead to isotropic collisionless components.

11.12.2 Gravitational-Wave Diagnostics

Mode-Dependent Damping Bounds

Different quasinormal modes impose distinct bounds on effective dissipation because each mode samples a different frequency band and damping sensitivity. Joint-mode fitting reduces degeneracy between source modeling and medium response by requiring a single constitutive parameter set to explain all modes simultaneously.

Event Stacking Strategy

Stacked posterior analysis across many events constrains the tails of (η, τ) more tightly than any single-event inference. Hierarchical pooling reduces sensitivity to event-level noise and highlights persistent constitutive structure, especially in the high-frequency ringdown window.

High-Frequency Tail Tests

High-frequency tail behavior is the most sensitive regime for testing elastic-dominant versus viscous-dominant propagation, because viscous suppression scales strongly with the frequency–relaxation

structure. Any systematic excess damping in this window exerts immediate exclusion pressure on the admissible parameter manifold.

11.12.3 Cross-Channel Bayesian Fusion

Likelihood Construction

A hierarchical likelihood fuses galaxy kinematics, morphology, lensing, and GW constraints under a shared constitutive parameterization. This suppresses channel-specific overfitting by requiring one parameter tuple to satisfy all observables simultaneously, which serves as the core non-contradiction test.

Prior Robustness Sweeps

Prior sweeps quantify the degree to which conclusions depend on astrophysical nuisance assumptions. Robust conclusions should persist across physically reasonable prior families, whereas fragile conclusions indicate hidden degeneracy or insufficiently informative data.

Posterior Geometry

Posterior geometry indicates whether a compact viable manifold exists or whether only disconnected, fine-tuned islands remain. Compact manifolds support model stability and transferability, whereas fragmented islands imply brittle calibration and weak predictive portability.

11.13 Implementation Handbook

This handbook provides operational standards so new contributors can reproduce, validate, and verify results without hidden workflow assumptions.

11.13.1 Workflow for New Researchers

Minimal Reproduction Path

The minimal path runs the baseline galaxy and GW notebooks, reproduces the key plots, and verifies equation-linked numeric tables. A complete minimum run should include environment fingerprinting, deterministic seed declaration, and a manifest of exact input datasets to guarantee replay fidelity.

Validation Protocol

Validation includes convergence tests, symbol–unit checks, and cross-file consistency checks between manuscript formulas and code implementations. Any mismatch between manuscript equations and executable code should be treated as a blocking error rather than as a documentation issue.

Failure Logging Standards

All failed runs should store traces, parameter snapshots, and diagnostic summaries to enable transparent debugging. Failure artifacts should include stack traces, solver tolerances, data hashes, and random-seed values so failures can be reproduced and triaged.

11.13.2 Equation-to-Code Traceability

Identifier Mapping

Each manuscript equation receives an identifier that maps to code functions and output artifacts. This map should be bidirectional: from manuscript to code and from code outputs back to manuscript

equation labels.

Unit Safety Layer

A unit-safety layer guards against silent dimensional mistakes in derived quantities such as G_{eff} and wake terms. Unit assertions should be executable and version-controlled, with assertion failures halting pipeline publication.

Regression Guardrails

Automated regression checks flag drift in benchmark outputs after code or parameter updates. Drift thresholds should be predefined by observable class so expected numerical noise is separated from substantive model changes.

11.13.3 Publication-Grade Output Pipeline

Figure Reproducibility

Figures are generated from manifest-locked runs with deterministic style templates. The figure build process should embed metadata linking each panel to its run ID, code commit hash, and input-manifest checksum.

Table Consistency Checks

Numerical tables are re-derived directly from source outputs during build to avoid manual transcription errors. Table-generation scripts should act as the single source of truth and should emit machine-readable companion files.

Archival Packaging

Final archival bundles include the manuscript, manifests, raw outputs, and verification scripts needed for third-party replication. A complete archive should also include a replication README with step-by-step commands and expected checksum outputs for validation.

Bibliography

[1] D. Finkelstein and J. Rubinstein. Connection between spin, statistics, and kinks. *Journal of Mathematical Physics*, 9(11):1762–1779, 1968. doi: 10.1063/1.1664538.

[2] E. P. Gross. Structure of a quantized vortex in boson systems. *Nuovo Cimento*, 20:454–477, 1961. doi: 10.1007/BF02731494.

[3] L. P. Pitaevskii. Vortex lines in an imperfect bose gas. *Soviet Physics JETP*, 13:451–454, 1961.

[4] C. J. Pethick and H. Smith. *Bose-Einstein Condensation in Dilute Gases*. Cambridge University Press, 2 edition, 2008. doi: 10.1017/CBO9780511802850.

[5] N. Aghanim et al. Planck 2018 results. vi. cosmological parameters. *Astronomy and Astrophysics*, 641:A6, 2020. doi: 10.1051/0004-6361/201833910.

[6] S. Chandrasekhar. Dynamical friction. i. general considerations: the coefficient of dynamical friction. *Astrophysical Journal*, 97:255–262, 1943. doi: 10.1086/144517.

[7] T. W. B. Kibble. Topology of cosmic domains and strings. *Journal of Physics A*, 9(8):1387–1398, 1976. doi: 10.1088/0305-4470/9/8/029.

[8] W. H. Zurek. Cosmological experiments in superfluid helium? *Nature*, 317:505–508, 1985. doi: 10.1038/317505a0.

[9] S. S. McGaugh, F. Lelli, and J. M. Schombert. The sparc database. *Astronomical Journal*, 152(6): 157, 2016. doi: 10.3847/0004-6256/152/6/157.

[10] B. P. Abbott et al. Observation of gravitational waves from a binary black hole merger. *Physical Review Letters*, 116(6):061102, 2016. doi: 10.1103/PhysRevLett.116.061102.

[11] B. P. Abbott et al. Tests of general relativity with gw150914. *Physical Review Letters*, 116(22): 221101, 2016. doi: 10.1103/PhysRevLett.116.221101.

[12] R. Abbott et al. Gwtc-3: Compact binary coalescences observed by ligo and virgo during o3b. *arXiv preprint*, 2021.

[13] W. Israel. Nonstationary irreversible thermodynamics: A causal relativistic theory. *Annals of Physics*, 100(1-2):310–331, 1976. doi: 10.1016/0003-4916(76)90064-6.

[14] W. Israel and J. M. Stewart. Transient relativistic thermodynamics and kinetic theory. *Annals of Physics*, 118(2):341–372, 1979. doi: 10.1016/0003-4916(79)90130-1.